

Exercise: You are waiting in line at the express checkout of the grocery store. Three customers are waiting ahead of you and one customer is currently being served. The checkout time for an individual customer has an exponential distribution with mean of 20 seconds. What is ^{the} probability that you will still be waiting for service to begin two minutes from now?

Note that the exponential assumption allows me to calculate this prob. without knowing how long the person currently being served has spent already being served - memoryless property

$$T = \text{time until I start service (in minutes)} \\ = T_0 + T_1 + T_2 + T_3$$

T_0 = remaining service time for cust. currently being served

T_1 = service time for person 1 waiting

T_2 = " " " " 2 "

T_3 = " " " " 3 "

$$E(T_i) = \frac{1}{3}$$

$$T_0, T_1, T_2, T_3 \sim \text{Exponential}(3) \text{ or } \text{Gamma}(1, 3)^\alpha$$

If I assume T_0, T_1, T_2, T_3 independent
then $T \sim \text{Gamma}(4, 3)$

1 The Normal(μ, σ^2) Distribution

2 The random variable X has the **Normal(μ, σ^2) Distribution** if, for $\sigma > 0$,

$$3 \quad f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad -\infty < x < \infty$$

4 This is sometimes called the **Gaussian Distribution**.

5 The case where $\mu = 0$ and $\sigma = 1$ is called the **standard normal distribution**.

6 The CDF of the standard normal distribution is commonly denoted with
7 Φ .

8 **Properties worth noting:** If $X \sim N(\mu, \sigma^2)$, then

9 1. $E(X) = \mu$

10 2. $V(X) = \sigma^2$

11 3. For $a \neq 0$ and any b , the random variable $aX + b$ has the $N(a\mu + b, a^2\sigma^2)$
12 distribution. Note that this implies that $(X - \mu)/\sigma$ has the standard
13 normal distribution. $a = 1/\sigma, b = -\mu/\sigma$

14 4. Linear combinations of independent normal random variables are also
15 normally distributed.

Example: Suppose that $\log(S)$ has the $\text{Normal}(\mu, \sigma^2)$ distribution. Let K be a positive constant. Then it holds that

$$E((S - K)^+) = \exp(\mu + \sigma^2/2) \Phi\left(\frac{\mu + \sigma^2 - \log(K)}{\sigma}\right) - K \Phi\left(\frac{\mu - \log K}{\sigma}\right).$$

(This can be thought of as the expected value of a call option with strike price K , with S representing the price of the underlying stock on the maturity date.)

Proof: Denote $Y = \log(S)$. Hence, Y is $N(\mu, \sigma^2)$ and we can see that

$$\begin{aligned} E((S - K)^+) &= E((e^Y - K)^+) \\ &= \int_{-\infty}^{\infty} (e^y - K)^+ f_Y(y) dy \quad \text{by Rule for Lazy Statistician} \\ &= \int_{\log(K)}^{\infty} (e^y - K) f_Y(y) dy \quad \text{since } e^y < K \text{ when } y < \log(K) \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K \int_{\log(K)}^{\infty} f_Y(y) dy \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K P(Y > \log(K)) \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K P\left(Z > \frac{\log(K) - \mu}{\sigma}\right) \\ &\quad \text{where } Z \text{ has the standard normal distribution} \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K \Phi\left(\frac{\mu - \log(K)}{\sigma}\right) \\ &\quad \text{using the fact that } \Phi(x) = 1 - \Phi(-x) \end{aligned}$$

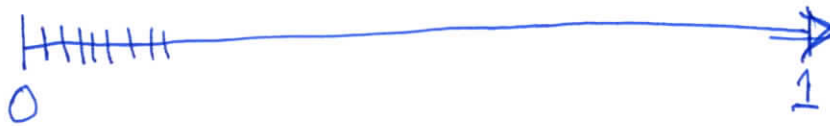
The remaining integral is tedious, but not difficult. It requires “completing the square” in the numerator of the expression that results when the normal density is plugged in for f_Y .

$$\begin{aligned}
 \int_{\log(K)}^{\infty} e^y f_Y(y) dy &= \int_{\log(K)}^{\infty} \frac{e^y}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{e^y}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2 - 2\mu y + \mu^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2 - 2\mu y + \mu^2 - 2\sigma^2 y}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2 - 2y(\mu + \sigma^2) + \mu^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \times \exp\left(-\frac{y^2 - 2y(\mu + \sigma^2) + (\mu + \sigma^2)^2 - (\mu + \sigma^2)^2 + \mu^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y - (\mu + \sigma^2))^2}{2\sigma^2}\right) \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) dy \\
 &= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) P(X > \log(K)) \\
 &\quad \text{where } X \sim N(\mu + \sigma^2, \sigma^2) \\
 &= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) P\left(Z > \frac{\log(K) - \mu - \sigma^2}{\sigma}\right) \\
 &\quad \text{where } Z \text{ is standard normal} \\
 &= \exp\left(\mu + \frac{\sigma^2}{2}\right) \Phi\left(\frac{\mu + \sigma^2 - \log(K)}{\sigma}\right)
 \end{aligned}$$

Substituting in this expression leads to the desired result. $\square$

The Poisson Process

In our initial discussion of the $\text{Poisson}(\lambda)$ distribution, we noted how the Poisson can be derived from the limit of $\text{binomial}(n, \lambda/n)$ random variables. There we imagined the interval $(0, 1)$ divided into n subintervals.



Now, imagine the same construction, but with the time interval extended to indefinitely. This leads to the **Poisson Process with rate λ** .

If I say that “events occur as a Poisson Process with rate λ ,” then

1. If Y equals the number of events during an interval of time of length T , then Y has the $\text{Poisson}(\lambda T)$ distribution.
2. The times between events are random variables with the Exponential(λ) distribution.
3. The times between events are independent random variables.
4. The numbers of events in disjoint time intervals are independent random variables.
5. The waiting time for k events has the $\text{Gamma}(k, \lambda)$ distribution. (This is also called the Erlang distribution.)

Exercise: Show that Property (1) above implies Property (2).

Suppose that time 0 corresponds to an "event"

T = time of subsequent event

For $t > 0$,

$$F_T(t) = P(T \leq t)$$

$$= P(\text{at least one event during } (0, t])$$

$$= 1 - P(\text{no events during } (0, t])$$

$$= 1 - \frac{e^{-\lambda t} (\lambda t)^0}{0!}$$

since # of events during $(0, t]$ is $\text{Poisson}(\lambda t)$

$$= 1 - e^{-\lambda t}$$

So, by uniqueness of CDF, conclude
that $T \sim \text{Exponential}(\lambda)$

Exercise: Suppose that X_2 equals the number of “jumps” between now and maturity date for a call option. Assume that these jumps arrive as a Poisson process with rate λ (in units of day^{-1}). When a jump does occur, its size is normally distributed with mean zero and variance τ^2 . There is still an underlying normal distribution to model the “normal” fluctuations in the stock price on the log-scale. Let T denote the time to maturity (in days).

So, the model can be written as follows: If S is the price on the maturity date, then

$$S = \exp\left(X_1 + \sum_{i=1}^{X_2} Z_i\right)$$

where X_1 has the $\text{Normal}(\mu, \sigma^2)$ distribution, X_2 has the $\text{Poisson}(\lambda T)$ distribution, and the Z_i have the $\text{Normal}(0, \tau^2)$ distribution.

Let K be the strike price for the option. What is $E((S - K)^+)$? You can assume that all of the random variables are independent.

$$E((S - K)^+) = \sum_{j=0}^{\infty} E((S - K)^+ | X_2 = j) P(X_2 = j)$$

a version of the LFTP for expectations
when $X_2 = j$, $X_1 + \sum_{i=1}^{X_2} Z_i \sim \mathcal{N}(\mu, \sigma^2 + j\tau^2)$

(when U, W uncorrelated, $V(U + W) = V(U) + V(W)$)

$$V(aU) = a^2 V(U)$$

So, $E((S-k)^+ | X_2=j)$ can be

found using our previous expression

(on page 50). Now $\log S$ is

$N(\mu, \sigma^2 + j\tau^2)$ instead
of $N(\mu, \sigma^2)$

Plus
$$P(X_2=j) = \frac{e^{-\lambda T} (\lambda T)^j}{j!}$$

Since $X_2 \sim \text{Poisson}(\lambda T)$

Properties of Expectation

If X, Y are random variables, and a, b , and c are constants, then

1. $E(c) = c$

2. $E(cX) = cE(X)$

3. $E(X + Y) = E(X) + E(Y)$ **if** either

(a) X, Y are both nonnegative

$\infty - \infty$

(b) X, Y are both nonpositive

(c) At least one of X, Y has finite expected values

4. $E(aX + b) = aE(X) + b$

5. $V(aX + b) = a^2V(X)$

Exercise: Suppose that a box contains n balls, labelled with numbers from 1 to n . Draws are made **with replacement** from the box m times, and the label on each ball is recorded after each draw. (Assume each ball is equally likely to be drawn.) What is the expected number of unique balls among the m draws?

$$X_i = \begin{cases} 1, & \text{if ball } i \text{ is drawn at least once} \\ 0, & \text{if not} \end{cases}$$

$$i = 1, 2, \dots, n$$

$$\sum_{i=1}^n X_i = \# \text{ of unique balls}$$

$$E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) \quad \text{by linearity of expectation}$$

$$E(X_i) = P(X_i = 1) = P(\text{ball } i \text{ drawn at least once})$$

$$= 1 - P(\text{ball } i \text{ never drawn}) = 1 - \left(\frac{n-1}{n}\right)^m$$

$$E(\# \text{ of unique balls}) = n \left(1 - \left(\frac{n-1}{n}\right)^m\right)$$

Jensen's Inequality

Property (4) on the ~~previous~~ page shows that if $g(x) = ax + b$, then $E(g(X)) = g(E(X))$. Unfortunately, however, it is not generally true that $E(g(X)) = g(E(X))$. What can be said?

Jensen's Inequality gives that if $g(\cdot)$ is convex **on the support of the random variable** X , then

$$E(g(X)) \geq g(E(X)).$$



Likewise, if $g(\cdot)$ is concave **on the support of the random variable** X , then

$$E(g(X)) \leq g(E(X)).$$

Example 1: The function $g(x) = x^2$ is convex on the entire real line, so for any random variable X , $E(X^2) \geq E(X)^2$. (Note that $V(X) = E(X^2) - E(X)^2$.)

Example 2: The function $g(x) = \log(x)$ is concave on $(0, \infty)$, so if X is a positive random variable, $E(\log(X)) \leq \log(E(X))$.



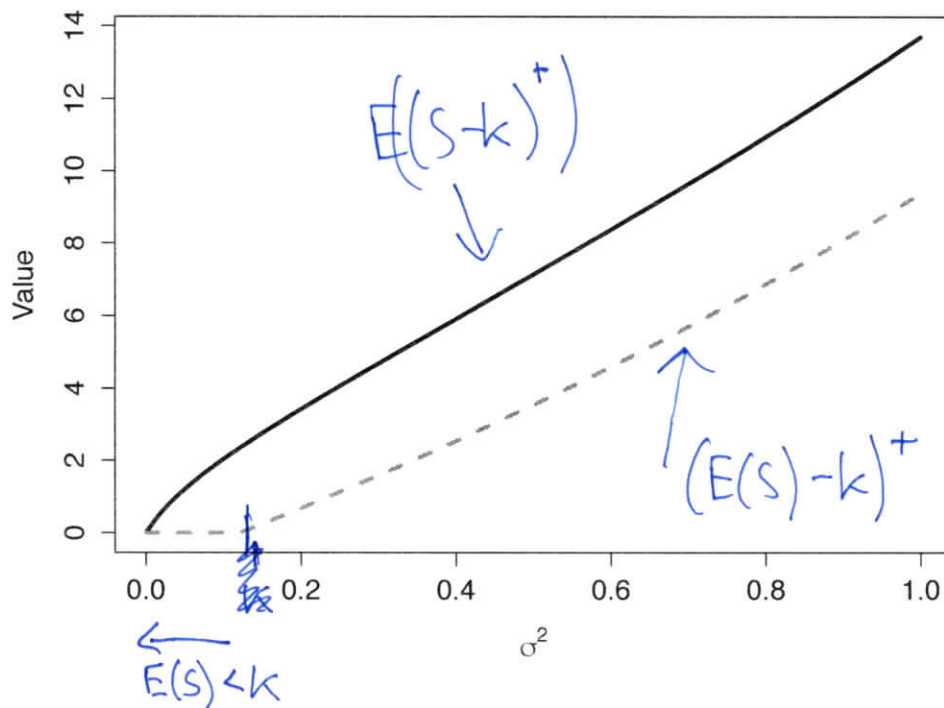
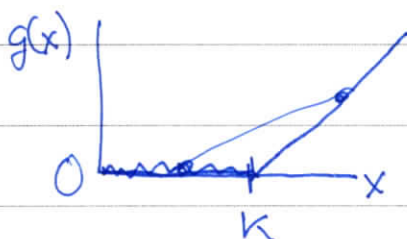


Figure 1: The case where $\mu = \log(16)$ and $K = 17$. One line shows the value of $E((S - K)^+)$, while the other is $(E(S) - K)^+$.

Exercise: Reconsider the exercise above in which $\log(S)$ was assumed to be $N(\mu, \sigma^2)$ and K is a positive constant. Figure 1 shows a case where $\mu = \log(16)$ and $K = 17$. The value of σ^2 varies along the horizontal axis. One line shows the value of $E((S - K)^+)$, while the other is $(E(S) - K)^+$.

Which one is which?

$$g(x) = (x - K)^+ \quad \uparrow \quad E(g(S)) \quad \uparrow \quad g(E(S))$$



convex, so $E(g(S)) \geq g(E(S))$

$$E((S - K)^+) \geq (E(S) - K)^+$$

Part 3: Multivariate Distributions

Exercise: Suppose that X and Y have pmf's

$$f_X(x) = \begin{cases} 0.5, & x = 100 \\ 0.5, & x = 250 \end{cases}$$

and

$$f_Y(y) = \begin{cases} 0.25, & y = 0 \\ 0.25, & y = 100 \\ 0.5, & y = 200 \end{cases}$$

What is $P(X > Y)$?

There is not enough info. to answer.

There could be dependence, need to model that to answer the question.

1 In the case of a pair of discrete random variables X and Y , we need to
2 specify the **joint probability mass function** in order questions such as the
3 previous one. We naturally write

4
$$f_{X,Y}(x, y) = P(X = x, Y = y) = \cancel{P(X=x)P(Y=y)}$$

5 Note that $f_{X,Y}(x, y) \geq 0$ for all x and y and that

6
$$\sum_x \sum_y f_{X,Y}(x, y) = 1$$

7 where the sum is over all pairs (x, y) for which $f_{X,Y}(x, y) > 0$.

8 The CDF is defined as

9
$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

10 The extension to more than two random variables is simple.

Exercise: A large insurance agency services a number of customers who have purchased both a homeowner's policy and an automobile policy from the agency. For each type of policy, a deductible amount must be specified. Let X be the deductible amount on the auto policy, and Y the deductible amount on the homeowner's policy. The joint pmf of (X, Y) is given by the following table:

x	y		
	0	100	200
100	.20	.10	.20
250	.05	.15	.30

What is the probability that X is larger than Y ?

$$P(X > Y) = \sum_{(x,y): x > y} f_{X,Y}(x,y)$$

$$= 0.20 + 0.05 + 0.15 + 0.30 = 0.70$$

The Continuous Case

The pair of random variables (X, Y) is **jointly continuous** with density $f_{X,Y}(x, y)$ if

1. $f_{X,Y}(x, y) \geq 0$ for all x, y

2.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

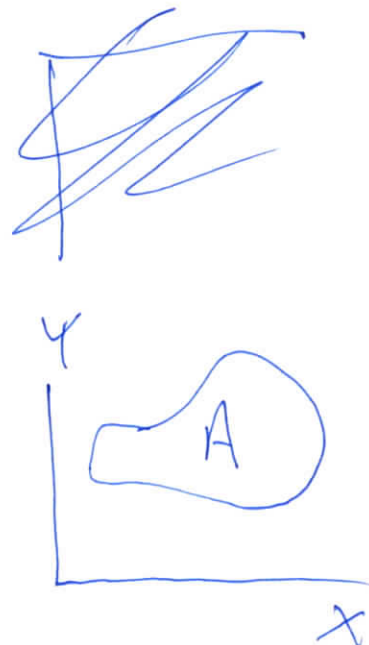
To calculate probabilities, integrate the pdf over the region of interest:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

Here, it is **not** the case that A need be rectangular.

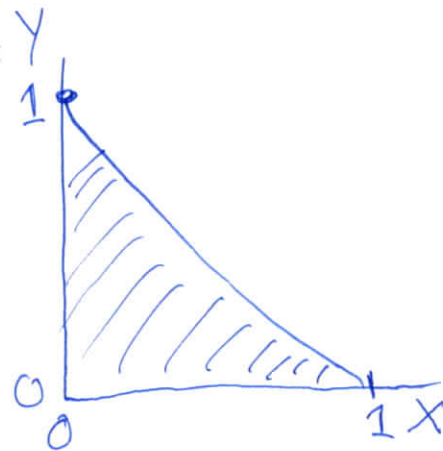
A useful property: In the continuous case,

$$\frac{\partial F_{X,Y}(x, y)}{\partial x \partial y} = f_{X,Y}(x, y)$$



Exercise: Suppose that the joint pdf of (X, Y) is given by:

$$f_{X,Y}(x, y) = \begin{cases} 24xy, & x > 0, y > 0 \\ & x + y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

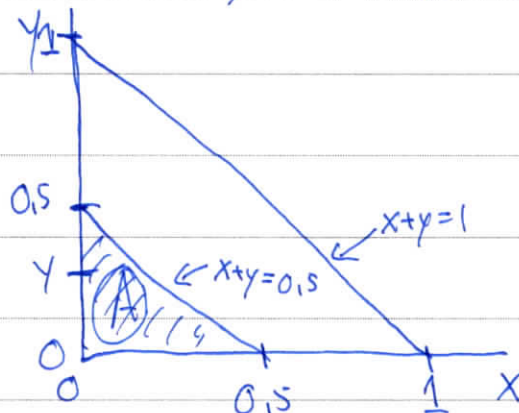


1. What is the probability that $X + Y$ is at most 0.5?

2. Find $P(X > Y)$.

3. Find the pdf of the random variable $T = X + Y$.

① $P(X+Y \leq 0.5) = P((X, Y) \in A)$



$$= \iint_A f_{X,Y}(x, y) dx dy$$

$$= \int_0^{0.5} \int_0^{0.5-y} 24xy dx dy = \frac{1}{16}$$

② 0.5 $P(X > Y)$

③ pdf of $T = X + Y$
Find F_T , take deriv.

For $0 < t < 1$,

$$\begin{aligned} P(T \leq t) &= P(X + Y \leq t) \\ &= \int_0^t \int_0^{t-y} 24xy \, dx \, dy = t^4 \end{aligned}$$

So, $F_T(t) = \begin{cases} 0, & t < 0 \\ t^4, & 0 \leq t \leq 1 \\ 1, & t > 1 \end{cases}$

So, $f_T(t) = \begin{cases} 4t^3, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$

1 The Beta(α, β) Distribution

2 In the previous Exercise, the random variable T has the Beta($\alpha = 4, \beta = 1$)
3 Distribution.

4 The random variable X has the **Beta(α, β) Distribution** if it has density

$$5 \quad f_X(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

6 This is a useful model for proportions or probabilities.

7 Note that $E(X) = \alpha/(\alpha + \beta)$.

8 Also note that if $\alpha = \beta = 1$, then the distribution is Uniform(0,1). Further,
9 if $\alpha = \beta$ then the distribution is symmetric about 0.5.

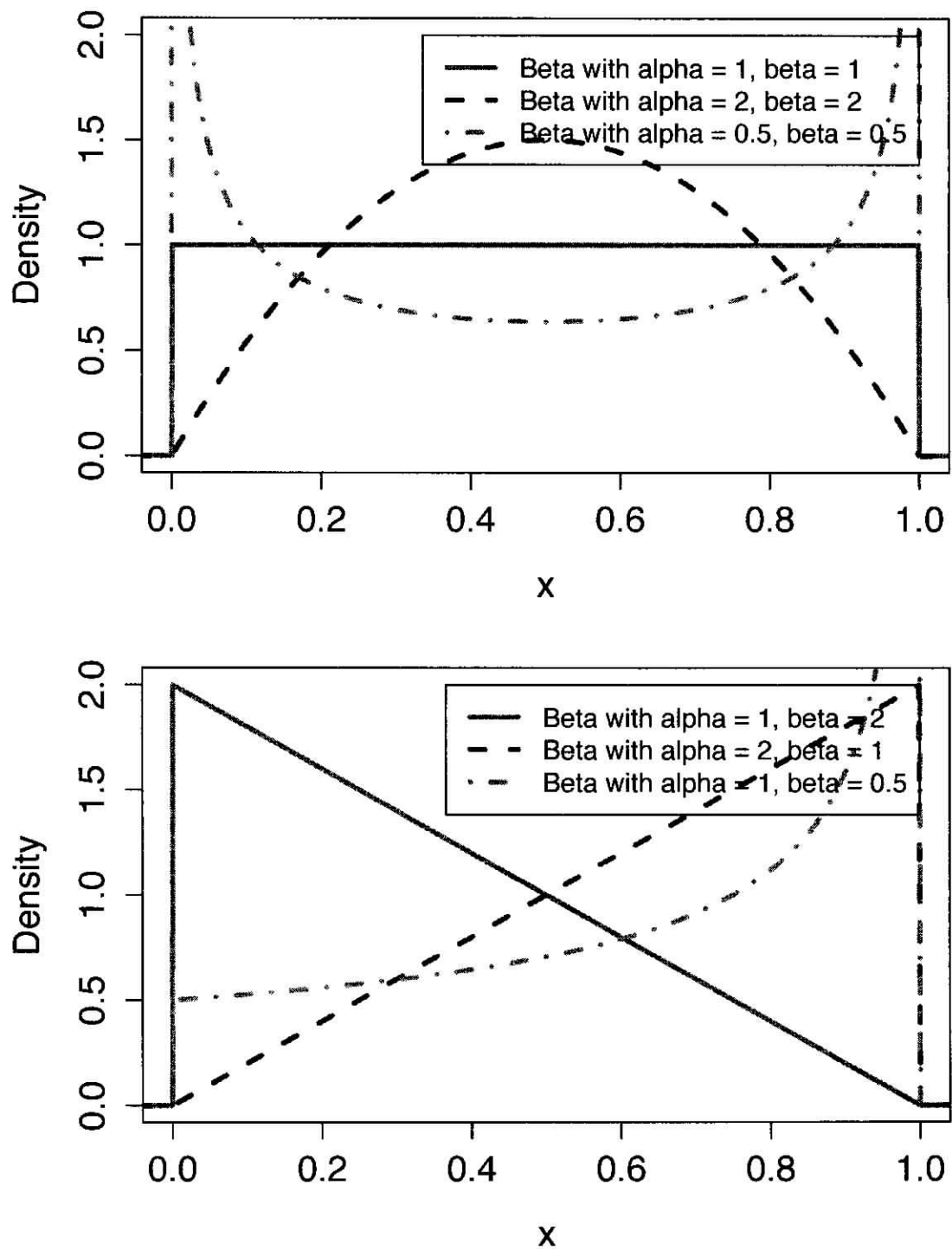


Figure 1: Beta Distribution pdfs.

Marginal Distributions

If one starts with the joint distribution (pmf or pdf) for (X, Y) , it is possible to derive the distribution for X ignoring Y and the distribution for Y ignoring X .

These are called the **marginal distributions** for X and Y , respectively.

These are simply the f_X and f_Y already described.

In the **discrete case**, you “sum over” the other random variable, i.e.,

$$f_X(x) = P(X = x) = \sum_y P(X = x, Y = y) = \sum_y f_{X,Y}(x, y)$$

Exercise: Find f_X and f_Y in the Insurance example above.

$$f_X(100) = \sum_{\substack{x \neq 100 \\ (x,y) \\ x=100}} f_{X,Y}(x,y) = 0,2 + 0,1 + 0,2 = 0,5$$

$$f_X(250) = 0,5$$

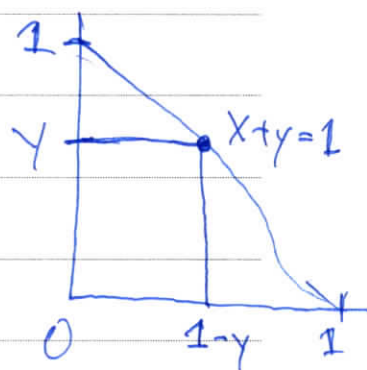
$$f_Y(0) = 0,25, \quad f_Y(100) = 0,25, \quad f_Y(200) = 0,50$$

In the **continuous case**, you “integrate out” the other variable:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

Exercise: Return to our “triangular support” example. Find f_X and f_Y .

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx \\ &= \int_0^{1-y} 24xy \, dx \end{aligned}$$



when $0 < y < 1$

$$= 12y(1-y)^2$$

$$\text{So, } f_Y(y) = \begin{cases} 12y(1-y)^2, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{ie. } Y \sim \text{Beta}(2, 3)$$

$$\text{Likewise, } X \sim \text{Beta}(2, 3)$$

Exercise: An electronic system has one each of two different types of components in joint operation. Let X and Y denote the random lengths of life of the components of type I and II, respectively. The joint function is:

$$f_{X,Y}(x,y) = \begin{cases} 8xe^{-2(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal pdf $f_X(x)$ and $f_Y(y)$.

$$f_X(x) = 4xe^{-2x}, \quad x > 0$$

$$X \sim \text{Gamma}(\alpha=2, \beta=2)$$

$$f_Y(y) = 2e^{-2y}, \quad y > 0$$

$$Y \sim \text{Exponential}(2)$$

1 In the above example, we can see that

2
$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \text{ for all } x, y$$

3 This implies that X and Y are **independent**.

4 The above can be taken as the definition of independent random variables
5 in both the discrete and continuous cases. Further, we say that $X_1, X_2, \dots, X_n$
6 are independent if for all $x_1, x_2, \dots, x_n$,

7
$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

1 Rule for the Lazy Statistician

For a general function $g(x, y)$, one can find expected values using

$$E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

Exercise: Try the case $g(x, y) = x$ to verify that the result is $E(X)$.

$$\begin{aligned} E(g(X, Y)) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right] dx \quad \left(\text{assume } E(X) < \infty \right) \\ &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= E(X) \end{aligned}$$